PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Introduction to Circuits Exam

Multiple Choice Questions (4 points each)

Instructions: Choose the best answer for each question.

Question 1: What is the main purpose of the lab described in the document?
A) To design a new type of circuit.
B) To introduce the concept of digital logic gates.
C) To reintroduce circuit concepts like the superposition principle.
D) To analyze the properties of semiconductors.

Question 2: According to the document, which instruments were used in the lab experiment?
A) Voltmeter, Ammeter, Ohmmeter
B) Signal generator, Oscilloscope, Power supply
C) Logic analyzer, Spectrum analyzer, Function generator
D) Multimeter, Breadboard, Soldering iron

Question 3: Based on the text, what frequency was supplied by the signal generator during the experiment when creating Table E1?
A) 1 kHz
B) 10 kHz
C) 20 kHz
D) 50 kHz

Question 4: In the circuit under test, what is the function of the capacitor in Figure 1?
A) To act as a short circuit.
B) To act as an open circuit.
C) To block DC signals.
D) To amplify AC signals.

Short Answer Questions (8 points each)

Instructions: Answer each question in 2-3 sentences.

Question 5: Explain the objective of the lab in determining the combined RMS output voltage value.

Question 6: State the superposition principle in your own words and explain its relevance to the experimental results.

Question 7: Describe a possible source of error that could account for discrepancies between simulated and experimental results, based on the text.

Problem-Solving Questions (12 points each)

Instructions: Provide detailed explanations for each answer.

Question 8: Based on the information in the tables, calculate the percent difference between the vO (rms) value in Table E1 and Table E3. Show your work.

Question 9: Describe how the circuit in Figure 1 is simplified for analysis when the capacitor is considered an open circuit (Figures 2a and 2b). What components are analyzed separately?

Question 10: Based on the description of C3, explain the purpose of simulating the circuit in Multisim. What specific elements of the simulated and experimental results are compared and why?